johnston-linear-matrix-algebra: residuals

equations measured 2026-09-05T12:23:12Z against the model of 2026-09-01

Corrected (8)

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
johnston-linear-matrix-algebra_ EQ0006 (was)	22	■ 0.141	\begin{array}{rlrl} & & \mathbf{x}+2(\mathbf{v}+ \mathbf{w}) & =-\mathbf{v}-3(\mathbf{x}-\mathbf{w}) \\ \Longrightarrow & \mathbf{x}+2 \mathbf{v}+2 \mathbf{w} & =-\mathbf{v}-3 \mathbf{x}+3 \mathbf{w} & \\ \Longrightarrow & 4 \mathbf{x} & =-3 \mathbf{v}+ \mathbf{w} & \\ \Longrightarrow & \mathbf{x} & =\frac{1}{4}(\mathbf{w}-3 \mathbf{v}) . & \\ \Longrightarrow & \text { (divide both sides by } 4) \end{array}	$\begin{array}{rlrl} & & \mathbf{x}+2(\mathbf{v}+\mathbf{w}) & =-\mathbf{v}-3(\mathbf{x}-\mathbf{w}) \\ \Longrightarrow & \mathbf{x}+2 \mathbf{v}+2 \mathbf{w} & =-\mathbf{v}-3 \mathbf{x}+3 \mathbf{w} & \\ \Longrightarrow & 4 \mathbf{x} & =-3 \mathbf{v}+\mathbf{w} & \\ \Longrightarrow & \mathbf{x} & =\frac{1}{4}(\mathbf{w}-3 \mathbf{v}) . & \\ \Longrightarrow & \text { (divide both sides by } 4) \end{array}$	—
johnston-linear-matrix-algebra_ EQ0006 (now)	22	—	\begin{array}{rll} & \mathbf{x}+2(\mathbf{v}+ \mathbf{w}) = -\mathbf{v}-3(\mathbf{x}-\mathbf{w}) & \\ \Longrightarrow & \mathbf{x}+2\mathbf{v}+2\mathbf{w} = -\mathbf{v}-3\mathbf{x}+3\mathbf{w} & \text{(expand parentheses)} \\ \Longrightarrow & 4\mathbf{x} = -3 \mathbf{v}+\mathbf{w} & \text{(add $3\mathbf{x}$, subtract $2\mathbf{v}+2\mathbf{w}$)} \\ \Longrightarrow & \mathbf{x} = \frac{1}{4}(\mathbf{w}-3\mathbf{v}). & \text{(divide both sides by 4)} \end{array}	*(not rendered)*	—
basis: inferred / ink					
johnston-linear-matrix-algebra_ EQ0014 (was)	27	■ 0.071	\left.\begin{array}{rl} (\mathbf{v}+\mathbf{w}) \cdot(\mathbf{x}+\mathbf{y}) & =(\mathbf{v}+\mathbf{w}) \cdot \mathbf{x}+(\mathbf{v}+\mathbf{w}) \cdot \mathbf{y} \\ & =\mathbf{x} \cdot(\mathbf{v}+\mathbf{w})+\mathbf{y} \cdot(\mathbf{v}+\mathbf{w}) \\ & =\mathbf{x} \cdot \mathbf{v}+\mathbf{x} \cdot \mathbf{w}+\mathbf{y} \cdot \mathbf{v}+\mathbf{y} \cdot \mathbf{w} \\ & =\mathbf{v} \cdot \mathbf{x}+\mathbf{w} \cdot \mathbf{x}+\mathbf{v} \cdot \mathbf{y}+\mathbf{w} \cdot \mathbf{y} \end{array}(\text { property (broperty (a) })\right)	$\left.\begin{array}{rl} (\mathbf{v}+\mathbf{w}) \cdot(\mathbf{x}+\mathbf{y}) & =(\mathbf{v}+\mathbf{w}) \cdot \mathbf{x}+(\mathbf{v}+\mathbf{w}) \cdot \mathbf{y} \\ & =\mathbf{x} \cdot(\mathbf{v}+\mathbf{w})+\mathbf{y} \cdot(\mathbf{v}+\mathbf{w}) \\ & =\mathbf{x} \cdot \mathbf{v}+\mathbf{x} \cdot \mathbf{w}+\mathbf{y} \cdot \mathbf{v}+\mathbf{y} \cdot \mathbf{w} \\ & =\mathbf{v} \cdot \mathbf{x}+\mathbf{w} \cdot \mathbf{x}+\mathbf{v} \cdot \mathbf{y}+\mathbf{w} \cdot \mathbf{y} \end{array}(\text { property (broperty (a) })\right)$	—
johnston-linear-matrix-algebra_ EQ0014 (now)	27	—	\begin{array}{rl} (\mathbf{v}+\mathbf{w}) \cdot(\mathbf{x}+\mathbf{y}) =(\mathbf{v}+\mathbf{w}) \cdot \mathbf{x}+(\mathbf{v}+\mathbf{w}) \cdot \mathbf{y} & (\text{property (b)})\\ = \mathbf{x} \cdot(\mathbf{v}+ \mathbf{w})+\mathbf{y} \cdot(\mathbf{v}+\mathbf{w}) & (\text{property (a)})\\ = \mathbf{x} \cdot \mathbf{v}+ \mathbf{x} \cdot \mathbf{w}+\mathbf{y} \cdot \mathbf{v}+ \mathbf{y} \cdot \mathbf{w} & (\text{property (b)})\\ = \mathbf{v} \cdot \mathbf{x}+\mathbf{w} \cdot \mathbf{x}+\mathbf{v} \cdot \mathbf{y}+\mathbf{w} \cdot \mathbf{y}. & (\text{property (a)}) \end{array}	$\begin{array}{rl} (\mathbf{v}+\mathbf{w}) \cdot(\mathbf{x}+\mathbf{y}) =(\mathbf{v}+\mathbf{w}) \cdot \mathbf{x}+(\mathbf{v}+\mathbf{w}) \cdot \mathbf{y} & (\text{property (b)})\\ = \mathbf{x} \cdot(\mathbf{v}+\mathbf{w})+\mathbf{y} \cdot(\mathbf{v}+\mathbf{w}) & (\text{property (a)})\\ = \mathbf{x} \cdot \mathbf{v}+\mathbf{x} \cdot \mathbf{w}+\mathbf{y} \cdot \mathbf{v}+\mathbf{y} \cdot \mathbf{w} & (\text{property (b)})\\ = \mathbf{v} \cdot \mathbf{x}+\mathbf{w} \cdot \mathbf{x}+\mathbf{v} \cdot \mathbf{y}+\mathbf{w} \cdot \mathbf{y}. & (\text{property (a)}) \end{array}$	—
basis: inferred / ink					
johnston-linear-matrix-algebra_ EQ0063 (was)	45	■ 0.136	\begin{aligned} A B & =\left[\begin{array}{ll} I_{3} & I_{3} \\ O & C \end{array}\right]\left[\begin{array}{ll} D & O \\ O & D \end{array}\right] \\ & \left.=\left[\begin{array}{ll} I_{3} & D+I_{3} \end{array}\right] \quad \text { O } \quad \begin{array}{ll} I_{3} & O+I_{3} \\ O & D+C \\ O \end{array} \quad \begin{array}{ll} O & O+C \\ O \end{array}\right]=\left[\begin{array}{cc} D & D \\ O & C D \end{array}\right] . \end{aligned}	$\begin{aligned} A B & =\left[\begin{array}{ll} I_{3} & I_{3} \\ O & C \end{array}\right]\left[\begin{array}{ll} D & O \\ O & D \end{array}\right] \\ & \left.=\left[\begin{array}{ll} I_{3} & D+I_{3} \end{array}\right] \quad \text { O } \quad \begin{array}{ll} I_{3} & O+I_{3} \\ O & D+C \\ O \end{array} \quad \begin{array}{ll} O & O+C \\ O \end{array}\right]=\left[\begin{array}{cc} D & D \\ O & C D \end{array}\right] . \end{aligned}$	—

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

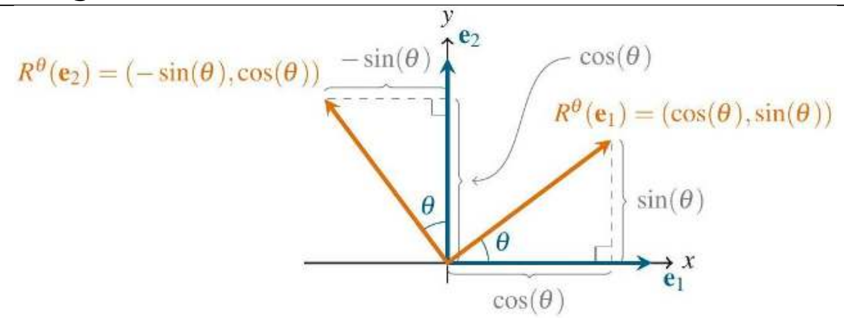
<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>johnston-linear-matrix-algebra_ EQ0063 (now)</td><td>45</td><td>—</td><td>\begin{aligned} \Longrightarrow \quad 4\mathbf{x} &= -3\mathbf{v} + \mathbf{w} \qquad &&\text{(add $3 \mathbf{x}$, subtract $2\mathbf{v}$)} \\ \Longrightarrow \quad \mathbf{x} &= \tfrac{1}{4}(\mathbf{w} - 3 \mathbf{v}). \qquad &&\text{(divide both sides by 4)} \end{aligned}</td><td>(not rendered)</td><td>—</td></tr>
<tr><td colspan="6">basis: inferred / ink</td></tr>
<tr><td>johnston-linear-matrix-algebra_ EQ0486 (was)</td><td>196</td><td>■ 0.011</td><td>\begin{array}{rrrrrrrr} v & + & w & + & x & + & y & - \\ v & - & 2 w & + & 2 x & + & 2 y & + \\ -v & + & w & + & x & & & + \\ & & & + & z & \\ 2 w & & - & x & + & y & - & 3 z \\ 2 v & & & - & 2 y & + & z & - \\ & & & & \end{array}</td><td>$\begin{array}{rrrrrrrr} v & + & w & + & x & + & y & - \\ v & - & 2 w & + & 2 x & + & 2 y & + \\ -v & + & w & + & x & & & + \\ & & & + & z & \\ 2 w & & - & x & + & y & - & 3 z \\ 2 v & & & - & 2 y & + & z & - \\ & & & & \end{array}$</td><td>—</td></tr>
<tr><td>johnston-linear-matrix-algebra_ EQ0486 (now)</td><td>196</td><td>—</td><td>\begin{array}{r c r c r c r c r c r} v & + & w & + & x & + & y & - & z & - & 1 \\ v & - & 2w & + & 2x & + & 2y & + & 2z & + & 2 \\ -v & + & w & + & x & & & + & z & & \\ & & 2w & - & x & + & y & - & 3z & + & 3 \\ 2v & + & w & & & - & 2y & + & z & - & 4 \\ \end{array}</td><td>$\begin{array}{r c r c r c r c r c r} v & + & w & + & x & + & y & - & z & - & 1 \\ v & - & 2w & + & 2x & + & 2y & + & 2z & + & 2 \\ -v & + & w & + & x & & & + & z & & \\ & & 2w & - & x & + & y & - & 3z & + & 3 \\ 2v & + & w & & & - & 2y & + & z & - & 4 \\ \end{array}$</td><td>—</td></tr>
<tr><td colspan="6">basis: inferred / ink</td></tr>
<tr><td>johnston-linear-matrix-algebra_ EQ0528 (was)</td><td>212</td><td>■ 0.001</td><td>\begin{array}{r} \\ \left.\xrightarrow{\begin{array}{c} z \\ R_{1} \end{array}} \begin{array}{c} S_{2} \quad S_{3} \\ R_{3}+R_{4} \\ R_{4}+R_{4} \\ \hline 0 \\ 0 \ \ 0 \end{array} \left\lvert\, \begin{array}{c|cccc|c} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ \hline \end{array}\left[\begin{array}{c|ccc|ccc} 1 & 0 & 0 & 1 & -2 & 1 & 0 \\ \hline 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1 & 1 & -2 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 1 \end{array}\right] \begin{array}{c} 2 \\ \text { pivot } \end{array}\right.\right] \end{array}</td><td>$\begin{array}{r} \\ \left.\xrightarrow{\begin{array}{c} z \\ R_{1} \end{array}} \begin{array}{c} S_{2} \quad S_{3} \\ R_{3}+R_{4} \\ R_{4}+R_{4} \\ \hline 0 \\ 0 \\ 0 \end{array} \left\lvert\, \begin{array}{c|cccc|c} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ \hline \end{array}\left[\begin{array}{c|ccc|ccc} 1 & 0 & 0 & 1 & -2 & 1 & 0 \\ \hline 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1 & 1 & -2 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 1 \end{array}\right] \begin{array}{c} 2 \\ \text { pivot } \end{array}\right.\right] \end{array}$</td><td>—</td></tr>
<tr><td>johnston-linear-matrix-algebra_ EQ0528 (now)</td><td>212</td><td>—</td><td>\xrightarrow{\begin{array}{c}z\\R_1\end{array}} \begin{array}{c}S_2\quad S_3\\R_3+R_4\\R_4+R_4\end{array} \left[\begin{array}{c|ccc|ccc|c}1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 2 \\ 0 & 0 & 1 & 0 & -1 & -1 & -1 & -1 \\ 0 & 0 & 0 & 1 & -2 & 1 & -1 & -2\end{array}\right]\curvearrowright\text{pivot here} \xrightarrow{\begin{array}{c}-R_4\\R_1-R_4\\R_3+R_4 \end{array}}\left[\begin{array}{c|ccc|ccc|c}1 & 0 & 0 & 1 & -2 & 1 & 0 & -2 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 2 \\ 0 & 0 & 1 & -1 & 1 & -2 & 0 & 1 \\ 0 & 0 & 0 & -1 & 2 & -1 & 1 & 2\end{array}\right]</td><td>$\xrightarrow{\begin{array}{c}z\\R_1\end{array}} \begin{array}{c}S_2\quad S_3\\R_3+R_4\\R_4+R_4\end{array} \left[\begin{array}{c|ccc|ccc|c}1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 2 \\ 0 & 0 & 1 & 0 & -1 & -1 & -1 & -1 \\ 0 & 0 & 0 & 1 & -2 & 1 & -1 & -2\end{array}\right]\curvearrowright\text{pivot here} \xrightarrow{\begin{array}{c}-R_4\\R_1-R_4\\R_3+R_4 \end{array}}\left[\begin{array}{c|ccc|ccc|c}1 & 0 & 0 & 1 & -2 & 1 & 0 & -2 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 2 \\ 0 & 0 & 1 & -1 & 1 & -2 & 0 & 1 \\ 0 & 0 & 0 & -1 & 2 & -1 & 1 & 2\end{array}\right]$</td><td>—</td></tr>
<tr><td colspan="6">basis: inferred / ink</td></tr>
<tr><td>johnston-linear-matrix-algebra_ EQ0543 (was)</td><td>218</td><td>■ 0.103</td><td>\begin{array}{rlrl} \operatorname{minimize}: & 3 y_{1}+3 y_{2}+3 y_{3}+3 y_{4}+3 y_{5} & \\ \text { subject to: } & y_{1}+y_{4}+y_{5} & \geq 1 \\ & y_{1}+y_{2}+y_{5} & \geq 1 \\ & y_{1}+y_{2}+y_{3} & \geq 1 \\ & y_{2}+y_{3}+y_{4} & \geq 1 \\ & y_{3}+y_{4}+y_{5} \geq 1 \\ y_{2}, \quad y_{3}, \quad y_{4}, \quad y_{5} & \geq 0 \end{array}</td><td>$\begin{array}{rlrl} \operatorname{minimize}: & 3 y_{1}+3 y_{2}+3 y_{3}+3 y_{4}+3 y_{5} & \\ \text { subject to: } & y_{1}+y_{4}+y_{5} & \geq 1 \\ & y_{1}+y_{2}+y_{5} & \geq 1 \\ & y_{1}+y_{2}+y_{3} & \geq 1 \\ & y_{2}+y_{3}+y_{4} & \geq 1 \\ & y_{3}+y_{4}+y_{5} \geq 1 \\ y_{2}, \quad y_{3}, \quad y_{4}, \quad y_{5} & \geq 0 \end{array}$</td><td>—</td></tr>
<tr><td colspan="6">Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured</td></tr>
</table>

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
johnston-linear-matrix-algebra EQ0543 (now)	218	—	`v = (6, 3, 3), (-2, 0, 6) = (8, 3, 0)`	$v = (6, 3, 3), (-2, 0, 6) = (8, 3, 0)$	—
basis: inferred / ink					
johnston-linear-matrix-algebra EQ0748 (was)	283	■ 0.116	`\begin{aligned} {\left[\begin{array}{lll} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{array}\right] } & \xrightarrow{\substack{R_{2}-R_{1} \\ R_{3}-R_{1}}} \left[\begin{array}{lll} 1 & 1 & 1 \\ 0 & 1 & 3 \ \ 0 & 2 & 8 \end{array}\right] \\ & \xrightarrow{ \substack{R_{1}-R_{2} \\ R_{3}-2 R_{2}}}\left[\begin{array}{ccc} 1 & 0 & -2 \\ 0 & 1 & 3 \\ 0 & 0 & 2 \end{array}\right] \\ \frac{\frac{1}{2} R_{3}}{}\left[\begin{array}{ccc} 1 & 0 & -2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{array}\right] & \xrightarrow{\substack{R_{1}+2 R_{3} \\ R_{2}-3 R_{3}}}\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right] . \end{aligned}`	$\begin{aligned} {\left[\begin{array}{lll} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{array}\right] } & \xrightarrow{\substack{R_{2}-R_{1} \\ R_{3}-R_{1}}} \left[\begin{array}{lll} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 2 & 8 \end{array}\right] \\ & \xrightarrow{ \substack{R_{1}-R_{2} \\ R_{3}-2 R_{2}}}\left[\begin{array}{ccc} 1 & 0 & -2 \\ 0 & 1 & 3 \\ 0 & 0 & 2 \end{array}\right] \\ \frac{\frac{1}{2} R_{3}}{}\left[\begin{array}{ccc} 1 & 0 & -2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{array}\right] & \xrightarrow{\substack{R_{1}+2 R_{3} \\ R_{2}-3 R_{3}}}\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right] . \end{aligned}$	—
johnston-linear-matrix-algebra EQ0748 (now)	283	—	`\mathbf{w} = (6,3,-3) - (-2,0,6) = (8,3,-9).`	$\mathbf{w} = (6,3,-3) - (-2,0,6) = (8,3,-9).$	—
basis: inferred / ink					
johnston-linear-matrix-algebra EQ1276 (was)	427	■ 0.022	`\left.3 x ^ { 2 } - 5 x - 2 \longdiv { \frac { x ^ { 2 } + 3 x + 1 } { 3 x ^ { 4 } + 4 x ^ { 3 } - 1 4 x ^ { 2 } - 1 1 x - 2 } }, \frac{3 x^{4}-5 x^{3}-2 x^{2}}{9 x^{3}-12 x^{2}-11 x-2}, 2, \frac{9 x^{3}-15 x^{2}-6 x}{3 x^{2}-5 x-2}, 3 \frac{3 x^{2}-5 x-2}{0}\right)`	$\left.3x^2-5x-2\right)\overline{\frac{x^2+3x+1}{3x^4+4x^3-14x^2-11x-2}}, \frac{3x^4-5x^3-2x^2}{9x^3-12x^2-11x-2}, 2, \frac{9x^3-15x^2-6x}{3x^2-5x-2}, 3\frac{3x^2-5x-2}{0}\Big)$	—
johnston-linear-matrix-algebra EQ1276 (now)	427	—	`\begin{array}{r} x^2+3x+1\phantom{xxxxxxxxx}\ \ 3x^2-5x-2\overline{)3x^4+4x^3-14x^2-11x-2}\\ \underline{3x^4-5x^3-2x^2\phantom{xxxxxxxxxx}}\\ 9x^3-12x^2-11x-2\\ \underline{9x^3-15x^2-6x\phantom{xxx}} \\ 3x^2-5x-2\\ \underline{3x^2-5x-2}\\ 0 \end{array}`	$\begin{array}{r} x^2+3x+1\\ 3x^2-5x-2\overline{)3x^4+4x^3-14x^2-11x-2}\\ \underline{3x^4-5x^3-2x^2}\\ 9x^3-12x^2-11x-2\\ \underline{9x^3-15x^2-6x} \\ 3x^2-5x-2\\ \underline{3x^2-5x-2}\\ 0 \end{array}$	—
basis: inferred / ink					

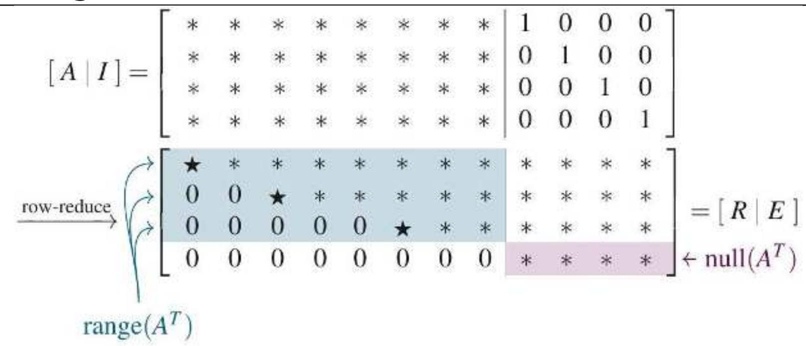
Conf. MathPix confidence: ■ $\geq$0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

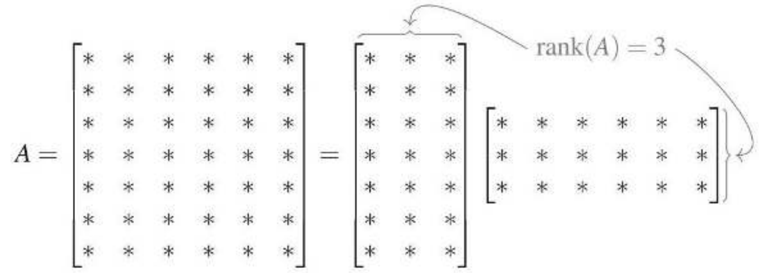
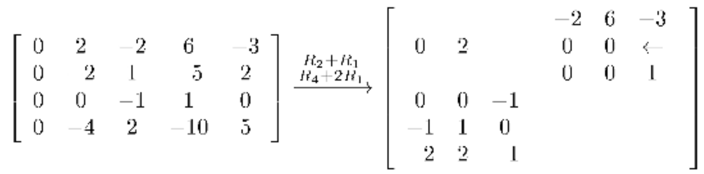
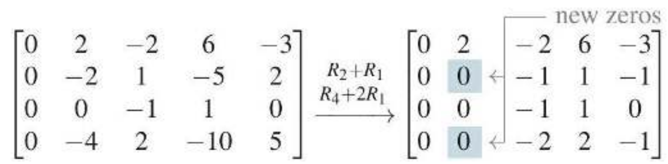
Unresolved

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra_EQ0114 **[conf 0.025]**	63	■ 0.025 •	\begin{figure} \[R^{\theta}\left(\mathbf{e}_{2} \right)=(-\sin (\theta), \cos (\theta)) \overbrace{ \cos (\theta)}^{-\sin (\theta)} \overbrace{ \mathbf{e}_{1}}^{\mathbf{e}_{2}} x \text { - } \underbrace{\cos (\theta)}_{R^{\theta}\left(\mathbf{e}_{1}\right)}=(\cos (\theta), \sin (\theta)) \] \captionsetup{labelformat=empty} \caption{Figure 1.19: \(R^{\theta}\) rotates the standard basis vectors \(\mathbf{e}_{1}\) and \(\mathbf{e}_{2}\) to \(R^{ \theta}\left(\mathbf{e}_{1}\right)=(\cos (\theta), \sin (\theta))\) and \(R^{\theta}\left(\mathbf{e}_{2} \right)=(-\sin (\theta), \cos (\theta))\), respectively.} \end{figure}	*(not rendered)*	
johnston-linear-matrix-algebra_EQ0947 **[conf 0.100]**	336	■ 0.100 •	\begin{array}{ll} \mathbf{p}_{1} \mathbf{q}_{1}^{*}= \frac{1}{3}\left[\begin{array}{l} 3 \\ 1 \\ 0 \end{array} \right]\left[\begin{array}{lll} 1 & 0 & -1 \end{array} \right]=\frac{1}{3}\left[\begin{array}{ccc} 3 & 0 & -3 \\ 1 & 0 & -1 \\ 0 & 0 & 0 \end{array}\right] & \text { pare } \\ \mathbf{p}_{2} \mathbf{q}_{2}^{*}= \frac{1}{3}\left[\begin{array}{l} 0 \\ 1 \\ 0 \end{array} \right]\left[\begin{array}{lll} -1 & 3 & 1 \end{array} \right]=\frac{1}{3}\left[\begin{array}{ccc} 0 & 0 & 0 \\ -1 & 3 & 1 \\ 0 & 0 & 0 \end{array}\right], & \text { and } \\ \mathbf{p}_{3} \mathbf{q}_{3}^{*}=\frac{1}{3} \left[\begin{array}{l} 1 \\ 0 \\ 1 \end{array}\right] \left[\begin{array}{lll} 0 & 0 & 3 \end{array}\right]= \frac{1}{3}\left[\begin{array}{lll} 0 & 0 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{array}\right] . & \end{array}	$\mathbf{p}_1\mathbf{q}_1^* = \frac{1}{3}\begin{bmatrix}3\\1\\0\end{bmatrix}\begin{bmatrix}1 & 0 & -1\end{bmatrix} = \frac{1}{3}\begin{bmatrix}3 & 0 & -3\\1 & 0 & -1\\0 & 0 & 0\end{bmatrix}$ pare $\mathbf{p}_2\mathbf{q}_2^* = \frac{1}{3}\begin{bmatrix}0\\1\\0\end{bmatrix}\begin{bmatrix}-1 & 3 & 1\end{bmatrix} = \frac{1}{3}\begin{bmatrix}0 & 0 & 0\\-1 & 3 & 1\\0 & 0 & 0\end{bmatrix},$ and $\mathbf{p}_3\mathbf{q}_3^* = \frac{1}{3}\begin{bmatrix}1\\0\\1\end{bmatrix}\begin{bmatrix}0 & 0 & 3\end{bmatrix} = \frac{1}{3}\begin{bmatrix}0 & 0 & 3\\0 & 0 & 0\\0 & 0 & 3\end{bmatrix}.$	$\mathbf{p}_1\mathbf{q}_1^* = \frac{1}{3}\begin{bmatrix}3\\1\\0\end{bmatrix}\begin{bmatrix}1 & 0 & -1\end{bmatrix} = \frac{1}{3}\begin{bmatrix}3 & 0 & -3\\1 & 0 & -1\\0 & 0 & 0\end{bmatrix}$ $\mathbf{p}_2\mathbf{q}_2^* = \frac{1}{3}\begin{bmatrix}0\\1\\0\end{bmatrix}\begin{bmatrix}-1 & 3 & 1\end{bmatrix} = \frac{1}{3}\begin{bmatrix}0 & 0 & 0\\-1 & 3 & 1\\0 & 0 & 0\end{bmatrix},$ and $\mathbf{p}_3\mathbf{q}_3^* = \frac{1}{3}\begin{bmatrix}1\\0\\1\end{bmatrix}\begin{bmatrix}0 & 0 & 3\end{bmatrix} = \frac{1}{3}\begin{bmatrix}0 & 0 & 3\\0 & 0 & 0\\0 & 0 & 3\end{bmatrix}.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra_EQ0439	170	■ 0.131 •	\begin{figure} \[\begin{aligned} {[A \mid I] } & =\left[\begin{array}{cccccccc|cccc} * & * & * & * & * & * & * & * & 1 & 0 & 0 & 0 \\ * & * & * & * & * & * & * & * & 0 & 1 & 0 & 0 \\ * & * & * & * & * & * & * & * & 0 & 0 & 1 & 0 \\ * & * & * & * & * & * & * & * & 0 & 0 & 0 & 1 \end{array}\right] \\ \xrightarrow{\text { row-reduce }} & (t)\left[\begin{array}{lllllll|llll} \star & * & * & * & * & * & * & * & * & * & * \\ 0 & 0 & \star & * & * & * & * & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & \star & * & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & * & * & * & * \end{array}\right]=[R \mid E] \leftarrow \operatorname{null}\left(A^{T}\right) \\ & \operatorname{range}\left(A^{T}\right) \end{aligned} \] \captionsetup{labelformat=empty} \caption{Figure 2.22: After we row-reduce \([A \mid I]\) to a row echelon form \([R \mid E]\), we can immediately read off bases of \(\operatorname{range}\left(A^{T}\right)\) and \(\operatorname{null}\left(A^{T}\right)\) from the rows of that matrix.} \end{figure}	*(not rendered)*	
johnston-linear-matrix-algebra_EQ0204	98	■ 0.137 •	\begin{aligned} \text { (equation 3) } & \begin{aligned} 2 x+4 y+5 z & =12 \\ -2(\text { equation 1 }) & -2 x-6 y+4 z \end{aligned}=-10 \\ \hline \text { (new equation 3) } & -2 y+9 z=2 \end{aligned}	$\begin{aligned} \text{(equation 3)} & \begin{aligned} 2x+4y+5z &= 12 \\ -2(\text{ equation 1 }) & -2x-6y+4z \end{aligned} = -10 \\ \hline \text{(new equation 3)} & -2y+9z=2 \end{aligned}$	$\begin{array}{rr} \text{(equation 3)} & 2x+4y+5z=12 \\ -2\text{(equation 1)} & -2x-6y+4z=-10 \\ \hline \text{(new equation 3)} & -2y+9z=2 \end{array}$
johnston-linear-matrix-algebra_EQ1055	361	■ 0.311 •	\begin{array}{lll} \left(\begin{array}{llll} 1 & 2 & 3 \end{array}\right) & \left(\begin{array}{llll} 2 & 1 & 3 \end{array}\right) & \left(\begin{array}{lll} 3 & 1 & 2 \end{array}\right) \\ \left(\begin{array}{lll} 1 & 3 & 2 \end{array}\right) & \left(\begin{array}{llll} 2 & 3 & 1 \end{array}\right) & \left(\begin{array}{llll} 3 & 2 & 1 \end{array}\right) \end{array}	$\begin{array}{lll} \left(\begin{array}{lll} 1 & 2 & 3 \end{array}\right) & \left(\begin{array}{lll} 2 & 1 & 3 \end{array}\right) & \left(\begin{array}{lll} 3 & 1 & 2 \end{array}\right) \\ \left(\begin{array}{lll} 1 & 3 & 2 \end{array}\right) & \left(\begin{array}{lll} 2 & 3 & 1 \end{array}\right) & \left(\begin{array}{lll} 3 & 2 & 1 \end{array}\right) \end{array}$	$\begin{array}{lll} (1\,2\,3) & (2\,1\,3) & (3\,1\,2) \\ (1\,3\,2) & (2\,3\,1) & (3\,2\,1). \end{array}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra EQ0575	224	■ 0.324 • C|+93	\begin{figure} \[\left.A=\left[\begin{array}{llllll} * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \end{array}\right]=\left[\begin{array}{lll} * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \end{array}\right] \left[\begin{array}{cccccc} * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \end{array} \right]\right\} \] \captionsetup{labelformat=empty} \caption{Figure 2.39: Theorem 2.C. 1 says that every matrix can be written as a product of a tall and skinny matrix and a short and fat matrix. The rank determines exactly how skinny and short the matrices in the product can be (as shown here, \(A\) has rank 3, so it can be written as a product of a matrix \(C\) with 3 columns and a matrix \(R\) with 3 rows).} \end{figure}	*(not rendered)*	
johnston-linear-matrix-algebra EQ0218	102	■ 0.407 • U|-9	\left[\begin{array}{ccccc} 0 & 2 & -2 & 6 & -3 \\ 0 & -2 & 1 & -5 & 2 \\ 0 & 0 & -1 & 1 & 0 \\ 0 & -4 & 2 & -10 & 5 \end{array}\right] \xrightarrow{\substack{R_{2}+R_{1} \\ R_{4}+2 R_{1}}}\left[\begin{array}{cccc} 0 & 2 & & \begin{array}{ccc} -2 & 6 & -3 \\ 0 & 0 & \leftarrow \\ 0 & 0 & 1 \end{array} \\ 0 & 0 & -1 \\ -1 & 1 & 0 \\ -2 & 2 & -1 \end{array}\right]		

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Flagged, not acted on

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$\det(A)=\det(B)=2, \quad \text{and} \quad p_A(\lambda)=p_B(\lambda)=-\lambda^3+3\lambda+2.$</td></tr>
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● C|+817</td><td><code>\begin{aligned} A B & =\left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array}\right]\left[\begin{array}{lll} 5 & 6 & 7 \\ 8 & 9 & 10 \end{array}\right] \\ & =\left[\begin{array}{ccc} (1,2) \cdot(5,8) & (1,2) \cdot(6,9) & (1,2) \cdot(7,10) \\ (3,4) \cdot(5,8) & (3,4) \cdot(6,9) & (3,4) \cdot(7,10) \end{array}\right] \\ & =\left[\begin{array}{lll} 1 \times 5+2 \times 8 & 1 \times 6+2 \times 9 & 1 \times 7+2 \times 10 \\ 3 \times 5+4 \times 8 & 3 \times 6+4 \times 9 & 3 \times 7+4 \times 10 \end{array}\right] \\ & =\left[\begin{array}{lll} 21 & 24 & 27 \\ 47 & 54 & 61 \end{array}\right] . \end{aligned}</code></td><td>$\begin{aligned} AB &= \left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array}\right]\left[\begin{array}{lll} 5 & 6 & 7 \\ 8 & 9 & 10 \end{array}\right] \\ &= \left[\begin{array}{ccc} (1,2)\cdot(5,8) & (1,2)\cdot(6,9) & (1,2)\cdot(7,10) \\ (3,4)\cdot(5,8) & (3,4)\cdot(6,9) & (3,4)\cdot(7,10) \end{array}\right] \\ &= \left[\begin{array}{lll} 1\times5+2\times8 & 1\times6+2\times9 & 1\times7+2\times10 \\ 3\times5+4\times8 & 3\times6+4\times9 & 3\times7+4\times10 \end{array}\right] \\ &= \left[\begin{array}{lll} 21 & 24 & 27 \\ 47 & 54 & 61 \end{array}\right]. \end{aligned}$</td><td>$\begin{aligned} AB &= \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 5 & 6 & 7 \\ 8 & 9 & 10 \end{bmatrix} \\ &= \begin{bmatrix} (1,2)\cdot(5,8) & (1,2)\cdot(6,9) & (1,2)\cdot(7,10) \\ (3,4)\cdot(5,8) & (3,4)\cdot(6,9) & (3,4)\cdot(7,10) \end{bmatrix} \\ &= \begin{bmatrix} 1\times5+2\times8 & 1\times6+2\times9 & 1\times7+2\times10 \\ 3\times5+4\times8 & 3\times6+4\times9 & 3\times7+4\times10 \end{bmatrix} \\ &= \begin{bmatrix} 21 & 24 & 27 \\ 47 & 54 & 61 \end{bmatrix}. \end{aligned}$</td></tr>
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Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured</td></tr>
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C|+20</td><td><code>\begin{array}{r} {\left[\begin{array}{lll|lll} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 4 & 0 & 1 & 0 \\ 1 & 3 & 9 & 0 & 0 & 1 \end{array}\right] \xrightarrow{\substack{R_{2}-R_{1} \\ R_{3}-R_{1}}}\left[\begin{array}{lll|ccl} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 3 & -1 & 1 & 0 \\ 0 & 2 & 8 & -1 & 0 & 1 \end{array}\right]} \\ \xrightarrow{R_{1}-R_{2}} \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 2 & -1 & 0 \\ 0 & 1 & 3 & -1 & 1 & 0 \\ 0 & 0 & 2 & 1 & -2 & 1 \end{array}\right] \\ \xrightarrow{\frac{1}{2} R_{3}} \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 2 & -1 & 0 \\ 0 & 1 & 3 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 / 2 & -1 & 1 / 2 \end{array}\right] \\ \xrightarrow{R_{1}+2 R_{3}}\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & -3 & 1 \\ 0 & 1 & 0 & -5 / 2 & 4 & -3 / 2 \\ 0 & 0 & 1 & 1 / 2 & -1 & 1 / 2 \end{array}\right] . \end{array}</code></td><td>$\begin{array}{r} {\left[\begin{array}{lll|lll} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 4 & 0 & 1 & 0 \\ 1 & 3 & 9 & 0 & 0 & 1 \end{array}\right] \xrightarrow{\substack{R_{2}-R_{1} \\ R_{3}-R_{1}}}\left[\begin{array}{lll|ccl} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 3 & -1 & 1 & 0 \\ 0 & 2 & 8 & -1 & 0 & 1 \end{array}\right]} \\ \xrightarrow{R_{1}-R_{2}} \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 2 & -1 & 0 \\ 0 & 1 & 3 & -1 & 1 & 0 \\ 0 & 0 & 2 & 1 & -2 & 1 \end{array}\right] \\ \xrightarrow{\frac{1}{2} R_{3}} \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 2 & -1 & 0 \\ 0 & 1 & 3 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 / 2 & -1 & 1 / 2 \end{array}\right] \\ \xrightarrow{R_{1}+2 R_{3}}\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & -3 & 1 \\ 0 & 1 & 0 & -5 / 2 & 4 & -3 / 2 \\ 0 & 0 & 1 & 1 / 2 & -1 & 1 / 2 \end{array}\right] . \end{array}$</td><td>$\begin{array}{r} {\left[\begin{array}{lll|lll} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 4 & 0 & 1 & 0 \\ 1 & 3 & 9 & 0 & 0 & 1 \end{array}\right] \xrightarrow{\substack{R_{2}-R_{1} \\ R_{3}-R_{1}}}\left[\begin{array}{lll|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 3 & -1 & 1 & 0 \\ 0 & 2 & 8 & -1 & 0 & 1 \end{array}\right]} \\ \xrightarrow{\substack{R_{1}-R_{2} \\ R_{3}-2R_{2}}} \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 2 & -1 & 0 \\ 0 & 1 & 3 & -1 & 1 & 0 \\ 0 & 0 & 2 & 1 & -2 & 1 \end{array}\right] \\ \xrightarrow{\frac{1}{2} R_{3}} \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 2 & -1 & 0 \\ 0 & 1 & 3 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 / 2 & -1 & 1 / 2 \end{array}\right] \\ \xrightarrow{\substack{R_{1}+2 R_{3} \\ R_{2}-3R_{3}}}\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & -3 & 1 \\ 0 & 1 & 0 & -5 / 2 & 4 & -3 / 2 \\ 0 & 0 & 1 & 1 / 2 & -1 & 1 / 2 \end{array}\right] . \end{array}$</td></tr>
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Conf. MathPix confidence: ≥ 0.9 0.5–0.9 <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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<td>$$\left[\begin{array}{c|ccc|ccc|c} 1 & 0 & 2 & 0 & 0 & -3 & 3 & 4 \\ \hline 0 & 1 & 3 & 0 & 0 & -3 & 5 & 7 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 0 & 2 & -2 & 5 \end{array}\right]$$
5/2 is smallest (and only) positive ratio</td>
<td>most negative entry in top row
$$\left[\begin{array}{c|ccc|ccc|c} 1 & 0 & 2 & 0 & 0 & -3 & 3 & 4 \\ \hline 0 & 1 & 3 & 0 & 0 & -3 & 5 & 7 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 0 & 2 & -2 & 5 \end{array}\right]$$
5/2 is smallest (and only) positive ratio</td>
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<td>■ 0.827
● C|+26</td>
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<td>$$\begin{gathered} {\left[\begin{array}{ccc|c} 1 & 3 & -2 & 5 \\ 1 & 5 & -8 & 9 \\ 2 & 4 & 5 & 12 \end{array}\right] \xrightarrow{R_{2}-R_{1}}\left[\begin{array}{ccc|c} 1 & 3 & -2 & 5 \\ 0 & 2 & -6 & 4 \\ 2 & 4 & 5 & 12 \end{array}\right]} \\ \xrightarrow[R_{3}-2 R_{1}]{ }\left[\begin{array}{ccc|c} 1 & 3 & -2 & 5 \\ 0 & 2 & -6 & 4 \\ 0 & -2 & 9 & 2 \end{array}\right] \xrightarrow{R_{3}+R_{2}}\left[\begin{array}{ccc|c} 1 & 3 & -2 & 5 \\ 0 & 2 & -6 & 4 \\ 0 & 0 & 3 & 6 \end{array}\right] . \end{gathered}$$</td>
<td>$$\begin{gathered} \left[\begin{array}{ccc|c} 1 & 3 & -2 & 5 \\ 1 & 5 & -8 & 9 \\ 2 & 4 & 5 & 12 \end{array}\right] \xrightarrow{R_{2}-R_{1}}\left[\begin{array}{ccc|c} 1 & 3 & -2 & 5 \\ 0 & 2 & -6 & 4 \\ 2 & 4 & 5 & 12 \end{array}\right] \\ \xrightarrow[R_{3}-2 R_{1}]{ }\left[\begin{array}{ccc|c} 1 & 3 & -2 & 5 \\ 0 & 2 & -6 & 4 \\ 0 & -2 & 9 & 2 \end{array}\right] \xrightarrow{R_{3}+R_{2}}\left[\begin{array}{ccc|c} 1 & 3 & -2 & 5 \\ 0 & 2 & -6 & 4 \\ 0 & 0 & 3 & 6 \end{array}\right] . \end{gathered}$$</td>
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<td>■ 0.865
● C|+60</td>
<td><code>\begin{aligned} & x_{1}=\frac{\operatorname{det}\left(\left[\begin{array}{ll} b_{1} & a_{1,2} \\ b_{2} & a_{2,2} \end{array}\right]\right)}{\operatorname{det} \left(\left[\begin{array}{ll} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{array}\right]\right)}=\frac{b_{1} a_{2,2}-a_{1,2} b_{2}}{a_{1,1} a_{2,2}-a_{1,2} a_{2,1}} \text { and } \\ & x_{2}=\frac{\operatorname{det}\left(\left[\begin{array}{ll} a_{1,1} & b_{1} \\ a_{2,1} & b_{2} \end{array}\right]\right)}{\operatorname{det}\left(\left[\begin{array}{ll} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{array}\right]\right)}=\frac{a_{1,1} b_{2}-b_{1} a_{2,1}}{a_{1,1} a_{2,2}-a_{1,2} a_{2,1}} . \end{aligned}</code></td>
<td>$$\begin{aligned} & x_{1}=\frac{\operatorname{det}\left(\left[\begin{array}{ll} b_{1} & a_{1,2} \\ b_{2} & a_{2,2} \end{array}\right]\right)}{\operatorname{det} \left(\left[\begin{array}{ll} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{array}\right]\right)}=\frac{b_{1} a_{2,2}-a_{1,2} b_{2}}{a_{1,1} a_{2,2}-a_{1,2} a_{2,1}} \text { and } \\ & x_{2}=\frac{\operatorname{det}\left(\left[\begin{array}{ll} a_{1,1} & b_{1} \\ a_{2,1} & b_{2} \end{array}\right]\right)}{\operatorname{det}\left(\left[\begin{array}{ll} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{array}\right]\right)}=\frac{a_{1,1} b_{2}-b_{1} a_{2,1}}{a_{1,1} a_{2,2}-a_{1,2} a_{2,1}} . \end{aligned}$$</td>
<td>$$\begin{aligned} & x_{1}=\frac{\operatorname{det}\left(\left[\begin{array}{ll} b_{1} & a_{1,2} \\ b_{2} & a_{2,2} \end{array}\right]\right)}{\operatorname{det} \left(\left[\begin{array}{ll} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{array}\right]\right)}=\frac{b_{1} a_{2,2}-a_{1,2} b_{2}}{a_{1,1} a_{2,2}-a_{1,2} a_{2,1}} \text { and } \\ & x_{2}=\frac{\operatorname{det}\left(\left[\begin{array}{ll} a_{1,1} & b_{1} \\ a_{2,1} & b_{2} \end{array}\right]\right)}{\operatorname{det}\left(\left[\begin{array}{ll} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{array}\right]\right)}=\frac{a_{1,1} b_{2}-b_{1} a_{2,1}}{a_{1,1} a_{2,2}-a_{1,2} a_{2,1}} . \end{aligned}$$</td>
</tr>
<tr><td colspan="6">Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured</td></tr>
</table>